



steamaco

Sell Power.
Profitably.



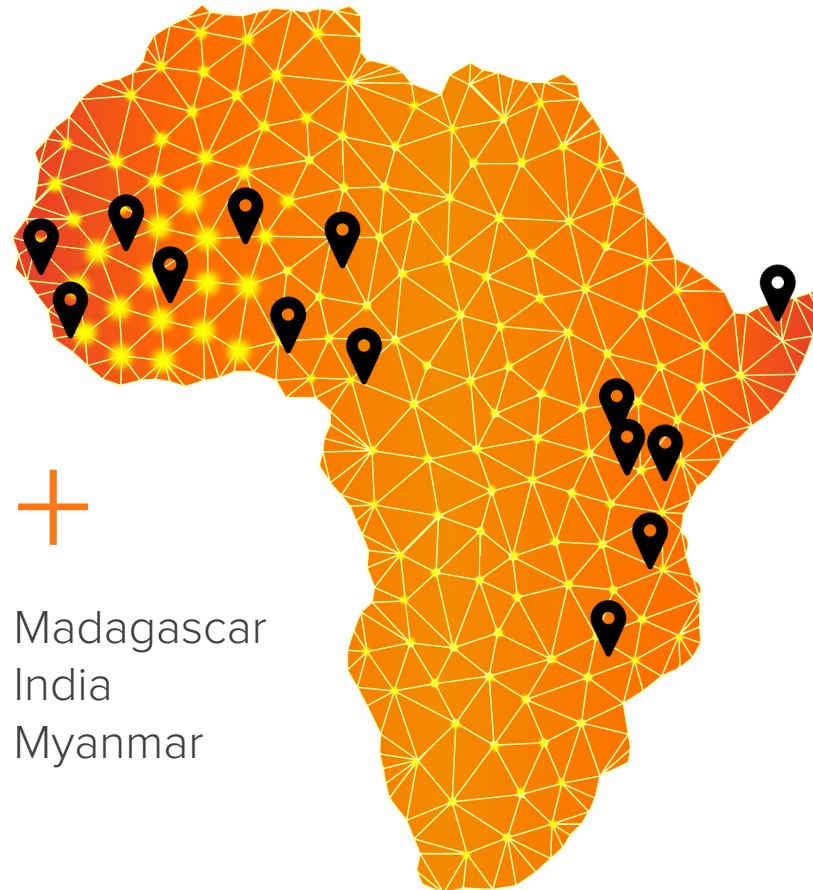
An aerial photograph of a rural village in a dry, hilly region. The village is composed of numerous small, simple houses with light-colored walls and flat roofs, scattered across a landscape of dry, brownish-yellow earth and sparse green trees. A prominent dirt road runs through the center of the village. In the foreground, a large, rectangular solar panel array is installed on a flat, open area, with a group of people gathered around it. The background shows rolling hills under a clear sky.

Our Vision: everyone has access to
clean, reliable and affordable power

Our Mission: help utilities
sell power profitably

Our presence:

Established tech partner to energy providers in Africa & Asia



Madagascar
India
Myanmar

30

Grid Operators

17

Countries

41k

Active Connections

96%

Average Uptime

Helping utilities deliver energy access with intelligence and automation

Steamaco's IoT products lets utilities deliver dynamic energy services that communities can rely on

Interoperable solution to guarantee energy service levels, earn revenue and build communities



Frictionless: API-led solution integrates with existing utility systems



Automation: tagged service delivery supports multiple energy service levels



Alerting: proactive maintenance and customer service



Visibility: complete accountability on usage and payments



Field-force management: improved efficiency of field service trips



Connectivity: low-cost, reliable and self-healing connection services



Protection: power quality monitoring to ensure grid health



Flexible: tariff choice and support for value added OTT⁽¹⁾ services



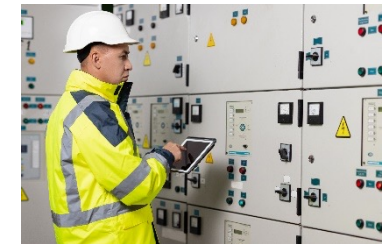
Trusted: better customer service through improved communication

Interfaces and support suitable for each use case

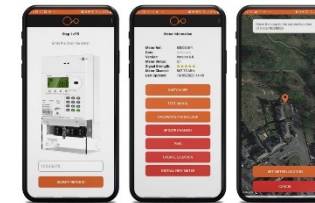
Simple user-interface



Utility HQ



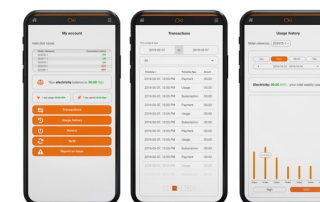
Managed workflows



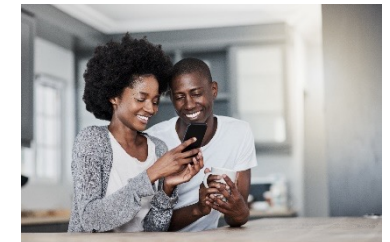
Utility Field Worker



Visibility on usage & costs



Utility customers



Delivering sustainable energy access projects with Nanogrid Power Hubs



Expanding access to energy in rural areas

The challenges in delivering sustainable energy access and essential social services

Rural energy access projects have a high failure rate

A significant proportion of mini grid projects fail soon after commissioning, due to:

- Lack of local acceptance
- Absence of local maintenance expertise
- Low energy consumption by rural customers

Factors contributing to failure

- Over investment in energy assets (mini grid instead of SHS)
- Lack of community engagement
- Failure to deliver reliable energy for community priorities
- Finite donor funded support
- Inability to pay for energy
- Focus on residential connections over productive uses

Access to Energy and Access to Social Services interlink

Expanding access to reliable healthcare & education services depends on access to sufficient and reliable energy:

- Rural brain-drain: Educated staff demand modern services and amenities, which require suitable energy services at both work and home
- Essential social services require refrigeration, heating, lighting, communications and clean water
- The energy transition requires the decarbonizing of local energy sources (i.e. displacing diesel generators) for:
 - Anchor load energy generation
 - Telecommunications towers
 - Mobility (e-scooter charging stations)
 - Productive use appliances (milling machines, oil presses, cookers, produce refrigerators etc.)

Solar Home Systems do not deliver sufficient energy for Anchor Loads, but the community cannot sustain a typically-sized mini grid asset without stimulation of energy demand & ability to pay

Unicef Case story: Energising Rural Healthcare in Malawi

Innovating a unique EaaS approach to deliver financially viable, community supported, nanogrids

Energy-as-a-Service model pioneered in Malawi
2020 to present

Funded with support by UKRI's Energy Catalyst program

- Partners

- **SteamaCo** – IoT smart meters and AMI platform
- **Balance of Storage (BOS)** – smart energy storage systems
- **Differ Community Power (DCP)** – Energy Service Provider (system integrator, financier, EPC, O&M, operator)

- EaaS Innovation: Nanogrid Power Hubs

- **Right-sized** solar generation and battery storage system to deliver guaranteed energy to healthcare facility (the **anchor load**)
- **Smart Nanogrid** used to distribute energy to connections based on priority of access

Objectives

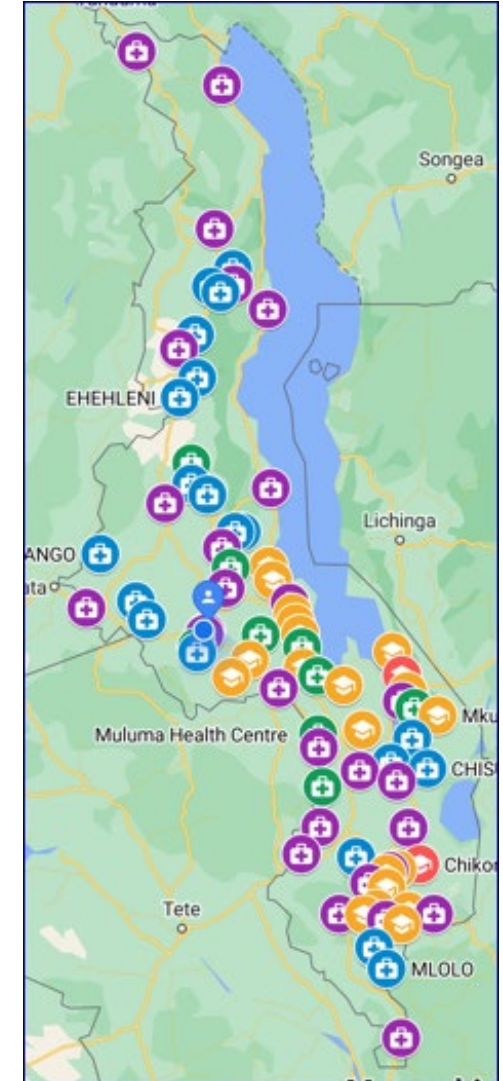
Partnering with Unicef Malawi to power 50 community healthcare centres

- Deliver guaranteed power for healthcare facilities and key appliances (e.g. vaccine fridges)
- Deliver clean water
- Energise staff homes

Project Statistics

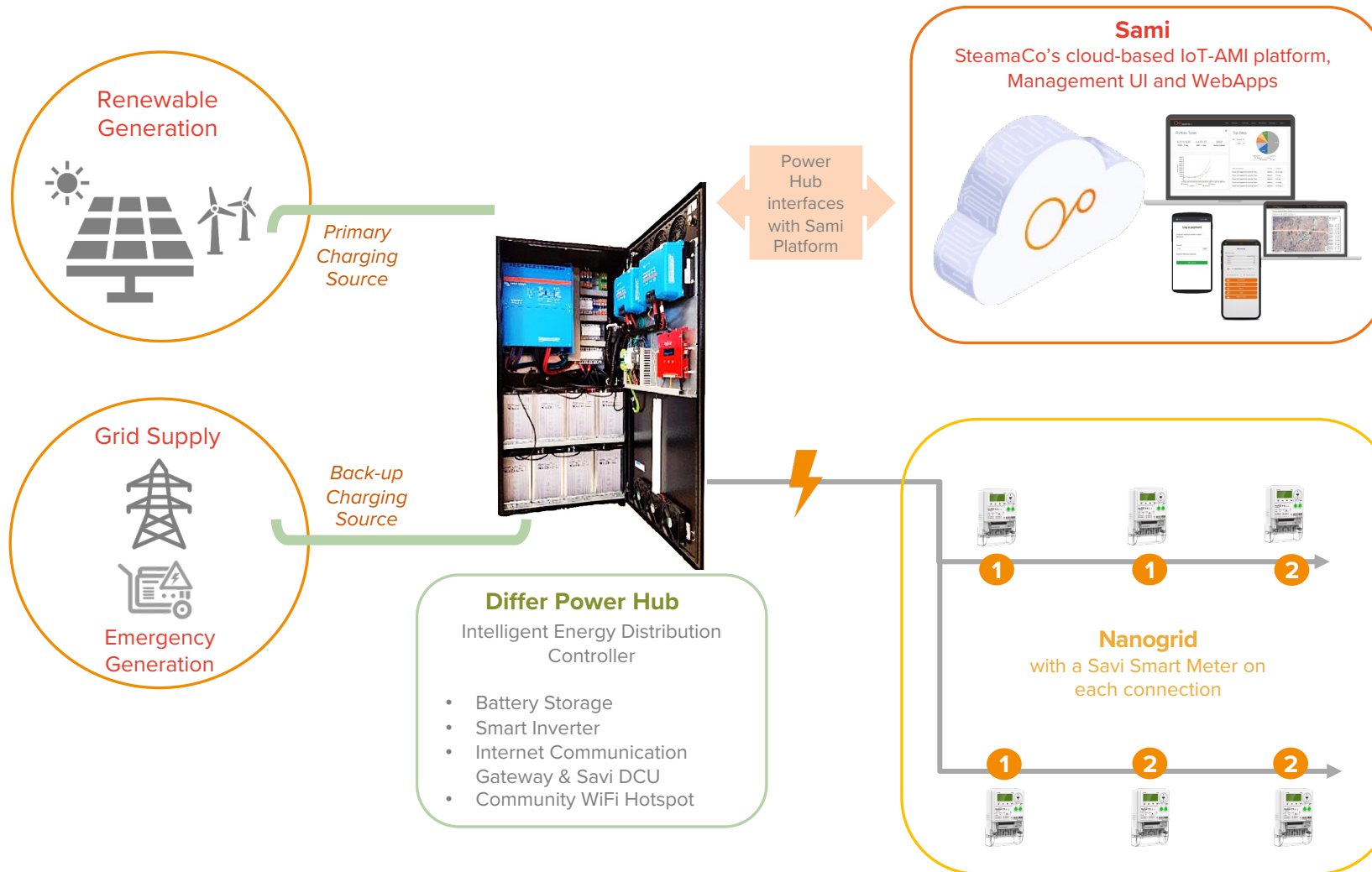
To date, the project has commenced operations at 46 sites, and is implementing remaining sites in Q322

- Solar PV units ranging from 4 – 30kW
- Battery storage sets ranging ~4 – 22kWh
- Between 4 and 10 connections managed by smart meters per site



Nanogrid Power Hubs : Guaranteed Energy for Productive Uses and Community Services

Delivering viable, sustainable energy access projects in healthcare, education and C&I applications



1

Essential Services & Anchor Tenant Connections

Anchor Load Connections

- Refrigeration & Heating Equipment
- Facility services (Water & Lights)
- Productive Users and Appliances

Guaranteed Energy Services

- Energy delivered 24/7 per agreed service level per connection
- **Paid-for Anchor loads deliver revenue assurance for Operator**
- Capacity to prioritise individual connections (e.g. fridges over lights)

2

Non-Essential Connections

Variable Service Connections

- Nearby residences or businesses
- Non-facility connections (area lighting, charging stations)
- Sami + Smart meters dynamically control power limits and supply switching

“When Available” Energy Services

- Excess available energy delivered on a variable service level agreement per connection
- **Additional revenue generation** for operator from selling excess available power, **or**
- **Subsidised power** for social benefits

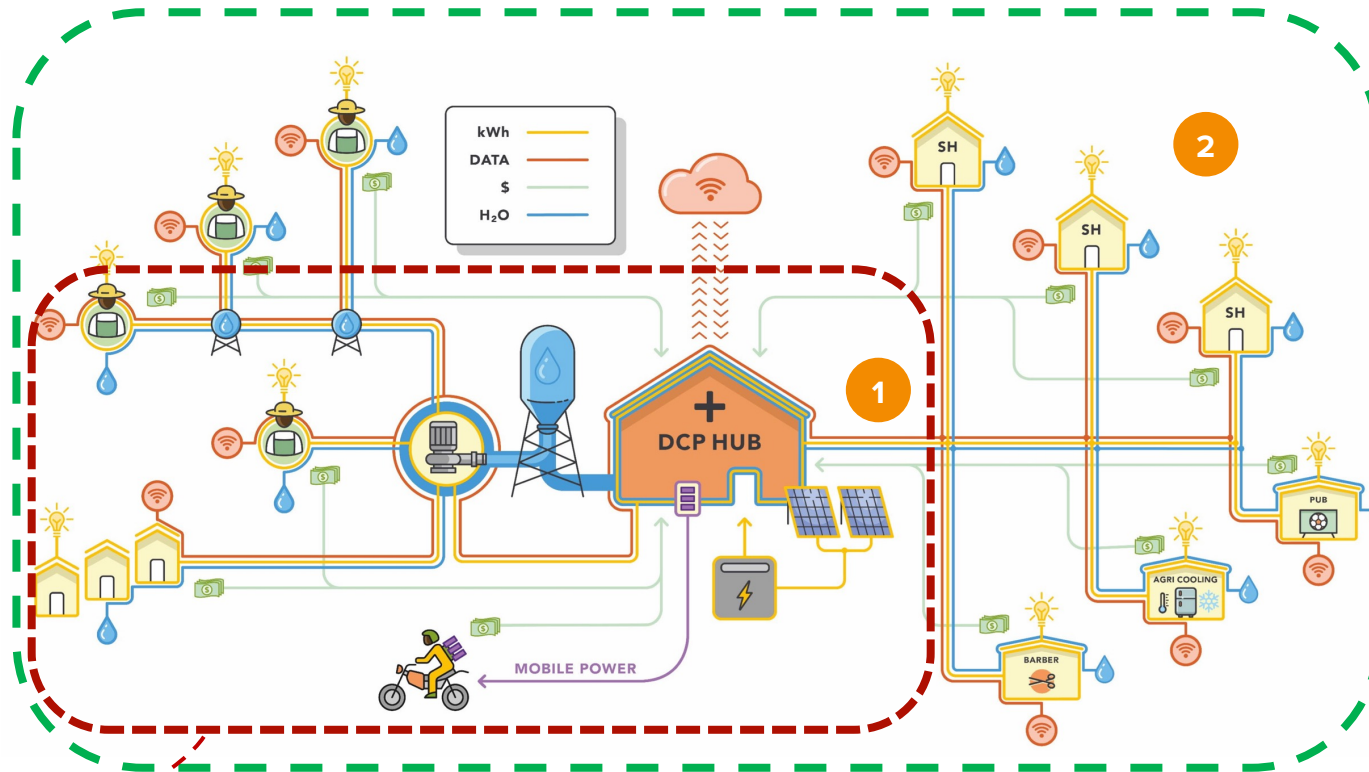
Unicef Case Story: Delivering energy access viably, centred on supplying anchor loads

Powering essential health services and stimulating demand with Nanogrid Power Hubs

1

Initial “right-sized” energy asset investment supporting the anchor load

- Deliver guaranteed power for healthcare facility and staff homes
- Each circuit controlled by a smart meter, e.g.:
 - Medical appliances
 - Water supply
 - Facility sockets & lights
 - Staff homes
 - Non-essential / productive appliances
- Nanogrid distributes excess energy to surrounding productive use appliances to earn revenue
- Variable service levels agreed for non-essential connections
- SHS devices in community residences



Ensuring Community Ownership

- Community engagement to prioritise energy services levels from outset
- Community employment created for O&M and Security roles
- Power Hub is recognised and valued as an essential community service

2

Financially sustainable energy asset expansion as community demand is stimulated

- Productive uses generate income for community
 - Increases ability to consume and pay for energy
 - Stimulates demand for energy
- Modular design
 - Genset & nanogrid grows incrementally to match increasing demand
- Long-term revenue assurance for ESP
 - Right-sized incremental investments
- Data analytics on usage and revenue define predicted exit path for donor-funding

Unicef Case Story: Project successes

Innovating management of energy services to deliver financially viable, community supported nanogrids

A Reliable, Right-Sized Energy Service

- ‘Guaranteed’ energy has been delivered at all times
 - Energy for prioritised connections always available
 - Battery storage capacity has ensured essential supply each night
- Operations
 - Battery banks typically recharge by noon each day
 - Significant excess capacity available during daylight hours
 - Proves the model: additional productive use loads can be accommodated across all sites
 - High demand appliances such as water pumps can operate during daytime without impacting supply to prioritised connections at night.

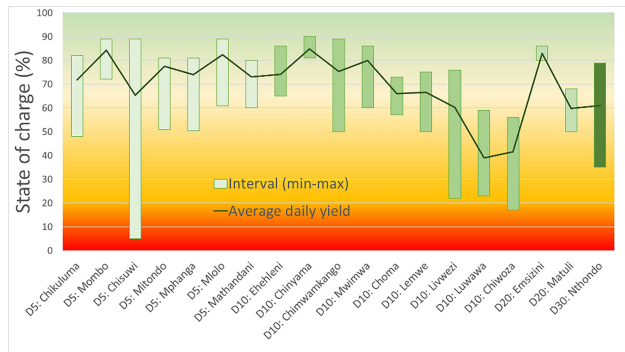


Figure 9: State of charge intervals for all sites

A Resilient Energy Service

- Energy services uninterrupted despite Storm Ana (Jan '22)
 - Every nanogrid delivered the guaranteed energy service levels despite very low rates of solar generation & battery charging during the multi-day storm.
- Minor impacts
 - Loss of mobile network at one site, and battery faults experienced did not interrupt energy supply to any prioritised connections
- Energy resilience for communities
 - Smart, distributed renewable energy assets ensure the lights stay on despite extreme weather events

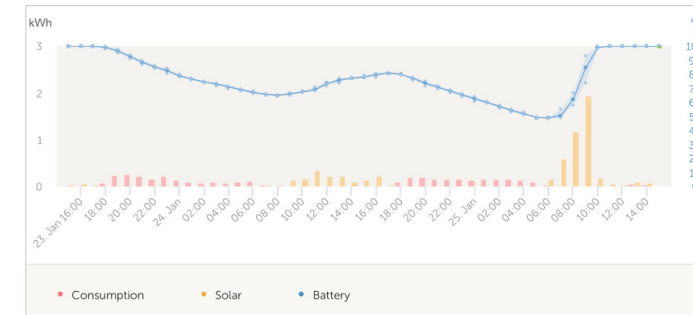


Figure 12: Power and state of charge at Chikuluma

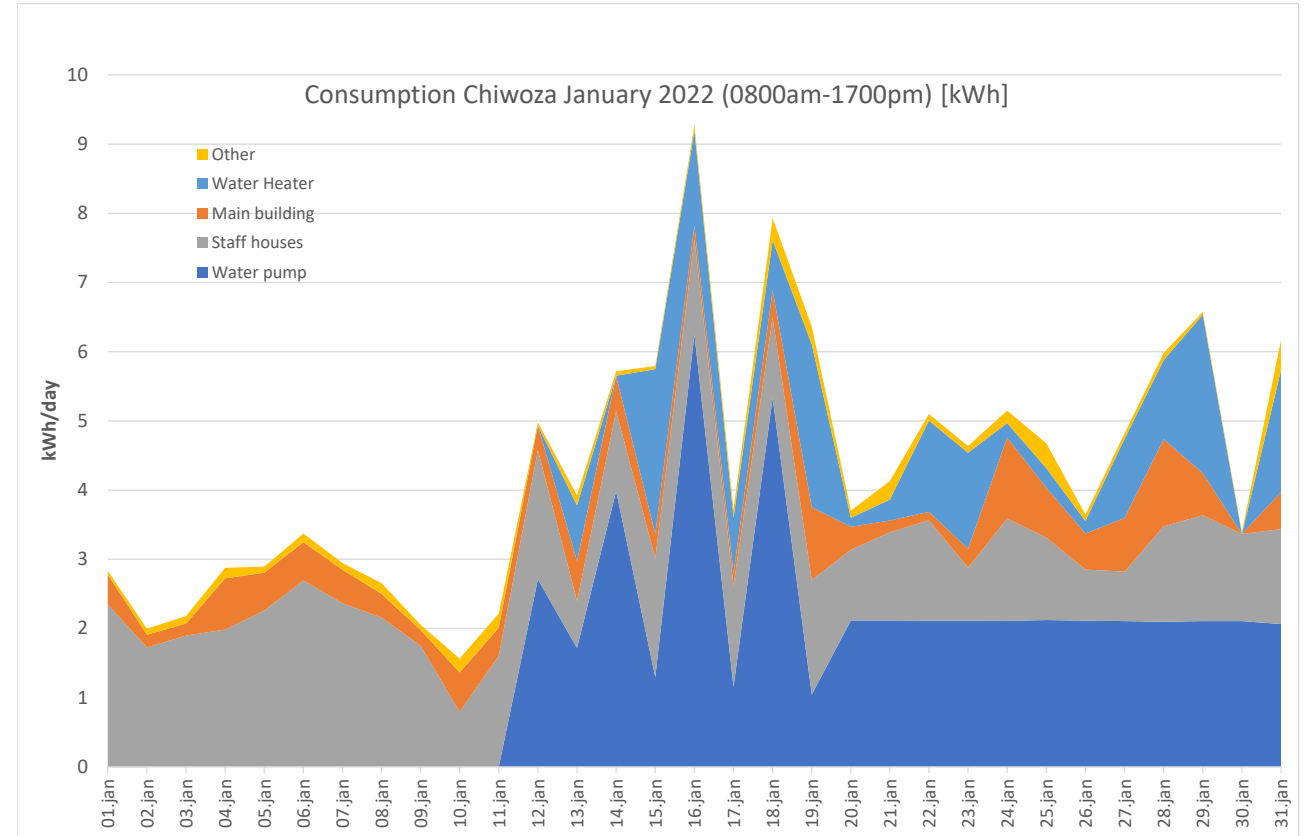
Unicef Case Story: Tagged Service Delivery

Detailed operational intelligence obtained from smart meters and inverters to control energy services

Digitised Nanogrids support variable service levels

Investment in smart meters and inverters and AMI software delivers results for nanogrids

- Innovative tagging of data points allows for detailed understanding of daily energy usage by the ESP
- Feedback mechanism from allows the dynamic and automated control of energy supply by the AMI & Power Hub, maintaining the guaranteed supply to prioritised connections:
 - Normal conditions: energy is supplied to every connection, according to agreed service levels, time-of-use-policies and power allowances
 - Battery charge status determines when non-essential supply needs to be curtailed or cut.
 - System can respond dynamically to non-normal situations (rainy, cloudy days, persistent bad weather)
- Individual connections can be prioritised according to service levels agreed with the community
- Consumers are kept informed by SMS or WebApp when energy services need to change in response to current load & storage conditions.



Nanogrid Power Hubs: Clean, reliable energy access, anywhere

Displacing diesel generators, growing businesses, developing communities

Wider applications are supported

- Supporting Productive Uses in areas lacking grid-supply
 - Nanogrid power hubs can be cost-efficiently deployed at any new energy access location besides for social services:
 - Small businesses & home industries
 - Rural markets and commercial plazas
 - Agri-businesses, water and irrigation infrastructure
- Urban and Rural grid supplied areas
 - Grid power is a backup charging source (when supply is available)
 - Displaces dirty diesel generators as primary source of guaranteed energy for homes, businesses, industrial operations
- Supports new Distribution Franchise concessions quickly
 - Allows for targeted, quick interventions to deliver energy services to essential services in franchise areas prior to any significant investment in supply augmentation
 - Provides much-needed operational and consumer intelligence that helps plan and de-risk large scale grid infrastructure upgrade investments



Any questions?

